

Executive Summary

Project Title: Pizza Sales Data Analysis using SQL

This project focuses on analyzing pizza sales data to uncover key business insights related to **sales performance, customer ordering behavior, revenue generation, and product popularity**. Using structured SQL queries, multiple datasets such as orders, order details, pizzas, and pizza types were integrated to perform comprehensive analysis.

The objective of this project was to help stakeholders understand **what sells best, when customers order most, which categories generate maximum revenue, and how sales grow over time**, enabling data-driven decision-making for business growth and operational efficiency.

Project Objectives

- Analyze total orders and overall revenue
- Identify top-performing pizzas by **price, quantity, and revenue**
- Understand customer ordering patterns by **time and date**
- Evaluate category-wise and size-wise pizza performance
- Track cumulative revenue trends over time
- Provide actionable insights and recommendations

Data Overview

The analysis was conducted using the following tables:

- **Orders** – Order date and time details
- **Order_Details** – Pizza-level order quantities
- **Pizzas** – Pizza sizes and prices
- **Pizza_Types** – Pizza names and categories

Key Insights & Findings

Order & Revenue Performance

- The total number of orders was calculated to understand overall demand.
- Total revenue was derived by multiplying pizza prices with quantities sold, giving a clear picture of business earnings.

Insight: The business generates strong revenue from repeat orders and high-volume pizza sales.

Product Pricing Analysis

- The **highest-priced pizza** was identified, helping understand premium offerings.

Insight: Premium pizzas contribute significantly to revenue, even if ordered less frequently.

Pizza Size Preference

- Analysis showed which pizza sizes are ordered most frequently.

Insight: Certain sizes dominate customer preference, indicating ideal portion demand.

Top-Selling Pizzas

- The **top 5 most ordered pizza types** were identified based on quantity.
- The **top 3 pizzas by revenue** were also identified.

Insight: High-quantity pizzas and high-revenue pizzas are not always the same—pricing plays a key role.

Category-Wise Performance

- Total quantity sold per pizza category (e.g., Classic, Veggie, Chicken).
- Revenue contribution percentage by category.

Insight: A few categories contribute the majority of total revenue, following the Pareto (80/20) principle.

Time-Based Ordering Behavior

- Orders were distributed by **hour of the day**.

Insight: Peak order hours indicate ideal timing for staffing, promotions, and inventory preparation.

Daily Order Trends

- Average number of pizzas ordered per day was calculated.

Insight: Provides a benchmark for daily sales expectations and operational planning.

Revenue Growth Over Time

- Cumulative revenue analysis showed how revenue builds day by day.

Insight: Sales trend analysis helps detect growth patterns, seasonality, and business stability.

Category-Level Revenue Leaders

- Top 3 pizzas by revenue were identified **within each category** using window functions.

Insight: Each category has star performers that should be prioritized in marketing.

Tools & Techniques Used

- **SQL Joins** (INNER JOIN)
 - **Aggregations** (SUM, COUNT, AVG)
 - **Window Functions** (RANK, OVER)
 - **Date & Time Functions**
 - **Subqueries**
 - **Revenue & Percentage Calculations**
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Business Recommendations

1. Focus on High-Revenue Pizzas

Promote top revenue-generating pizzas through:

- Featured menu placement
- Combo offers
- Upselling strategies

This will maximize profit without increasing operational costs.

2. Optimize Inventory & Staffing During Peak Hours

Since order volume peaks during specific hours:

- Increase staff availability during peak times
- Prepare ingredients in advance

Improves customer satisfaction and reduces wait time.

3. Promote High-Performing Categories

Categories contributing the highest revenue should:

- Get priority in advertisements
- Be included in discount campaigns carefully

Ensures revenue stability while increasing order volume.

4. Improve Low-Performing Pizzas

For pizzas with low sales:

- Review pricing
- Improve visibility
- Bundle with popular items

Helps reduce menu inefficiency.

5. Leverage Size Preferences

Since customers prefer specific pizza sizes:

- Focus promotions on preferred sizes
- Reduce inventory for low-demand sizes

Optimizes cost and reduces wastage.

6. Track Revenue Growth Regularly

Cumulative revenue analysis should be monitored monthly:

- Identify slow-growth periods
- Plan offers during low-demand days

Supports proactive business planning.

Conclusion

This SQL-based pizza sales analysis successfully transformed raw transactional data into **meaningful business insights**. The project demonstrates strong skills in **data analysis, SQL querying, problem-solving, and business understanding**. The insights derived can directly support **strategic planning, marketing optimization, and operational efficiency**.